Adam Shaw

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First author on three papers in scientific machine learning and representation learning, spanning gauge-equivariant neural networks, learned preconditioning for iterative solvers, and topological analysis of neural representations. Currently enrolled in a joint B.S./M.S. program with plans to apply to CS Ph.D. programs.

EDUCATION

University of Southern California — Los Angeles, CA

M.S. in Statistics — Expected December 2027
B.S. in Applied and Computational Mathematics, Minor in Computer Science — Expected May 2027

Undergraduate GPA 3.7 • Graduate GPA 4.0

University of Edinburgh (Semester Abroad) — Spring 2026

PUBLICATIONS

[1] **Adam Shaw**, Yixuan Sun, Srinivas Eswar, William Detmold, Phiala Shanahan, Xiaoye Sherry Li, and Yang Liu. "Learning Lattice Variables for Matrix-Free Wilson-Dirac Preconditioning in Lattice QCD." Accepted at the NeurIPS 2026 Workshop on Representations for the Physical Sciences.

[2] **Adam Shaw**, Jiayu Li, Michael Sperling, Michael Kim, and Alvin Jin. "Topological Simplification in Predictive Coding Networks." *Proceedings of the 2nd Conference on Topology, Algebra, and Geometry in Data Science (TAG-DS 2026), PMLR 334(2):316–332, 2026.*

[3] **Adam Shaw**, Jiayu Li, Michael Sperling, Michael Kim, and Alvin Jin. "Symmetry Acquisition in Predictive Coding Networks." *ICML 2026 Workshop on Weight-Space Symmetries.* Accepted (non-archival).

RESEARCH EXPERIENCE

Argonne National Laboratory (SULI Intern) — Summer 2026
Equivariant deep learning for large-scale numerical solvers — Lemont, IL

- Designed **matrix-free gauge-equivariant neural preconditioners** that learn to accelerate the iterative solution of the Dirac equation in lattice quantum chromodynamics (LQCD).
- Achieved **zero-shot transfer across lattice volumes**, with models trained on smaller lattices matching or exceeding their own speedup when run on larger ones, whereas the SOTA degraded without re-tuning.
- Built the full multi-GPU data-parallel training and evaluation pipeline in JAX, **training end-to-end** through **unrolled iterative solver iterations**.
- Proposed three novel architectures and diagnosed why two failed to improve results, tracing the outcome to specific structural properties and identifying directions for future improvement.

University of Southern California — Fall 2025 – Spring 2026

Topological analysis of learned representations in predictive coding networks — Los Angeles, CA

- Studied how **learned representations simplify layer-by-layer** in invertible neural networks and how that simplification affects a network's ability to invert; "simplification" was measured via topological invariants.
- Built an extensive Python research codebase implementing predictive coding networks in JAX and a representation-analysis stack in Ripser, NumPy, and scikit-learn.
- Produced two accepted papers and two conference posters, one chosen for a lightning talk (TAG-DS).

INDUSTRY EXPERIENCE

Google — Associate Software Developer Intern — Summer 2025

Workspace AI Platform — Gemini integration across Docs, Slides, Gmail, and more — New York, NY

- Designed and built an access control system for custom Gems (personalized Gemini instances), streamlining creation and secure sharing across Google Workspace teams.
- Implemented role-based permissions in C++ on Google's internal server framework, handling authorization and routing for millions of potential users.
- Automated enum ID generation across a 10,000+ line shared codebase in Python, eliminating merge conflicts for a multi-engineer team.

Rollick — Software Engineer Intern — Summer 2024

Front-end development and campaign personalization — Chicago, IL

- Built modular front-end templates in Nunjucks and jQuery and optimized UTM-based personalization, working in a standard review-and-merge Git workflow.

AWARDS AND HONORS

USC Presidential Scholarship — 2023

National Merit Scholarship Winner — 2023

TECHNICAL SKILLS

Languages	Python, C++, JavaScript
ML & scientific computing	JAX, NumPy, Pandas, scikit-learn, Ripser, Matplotlib
Methods	Equivariant and invertible neural architectures, automatic differentiation through iterative solvers, numerical linear algebra and Krylov solvers, persistent homology
Tools	Git/GitHub, Piper, Blaze, Protobuf, gRPC, HPC clusters (PBS), Linux, Bash, LaTeX

RELEVANT COURSEWORK

Undergraduate	Machine Learning, Mathematics of ML, NLP, Mathematical Statistics, Real Analysis
Graduate	Analysis of Algorithms, Numerical Methods, Applied Probability